Examiner only

9. All hydrocarbons can be burned but, apart from in combustion reactions, alkenes are more reactive than alkanes.

(a) Describe the bonding in propene and use this to explain its reactivity. [5]

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(b) Draw the mechanism for the reaction between propene and bromine. You should show any relevant dipoles, lone pairs of electrons and curly arrows to indicate the movement of pairs of electrons. [4]

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Turn over.

Examiner only

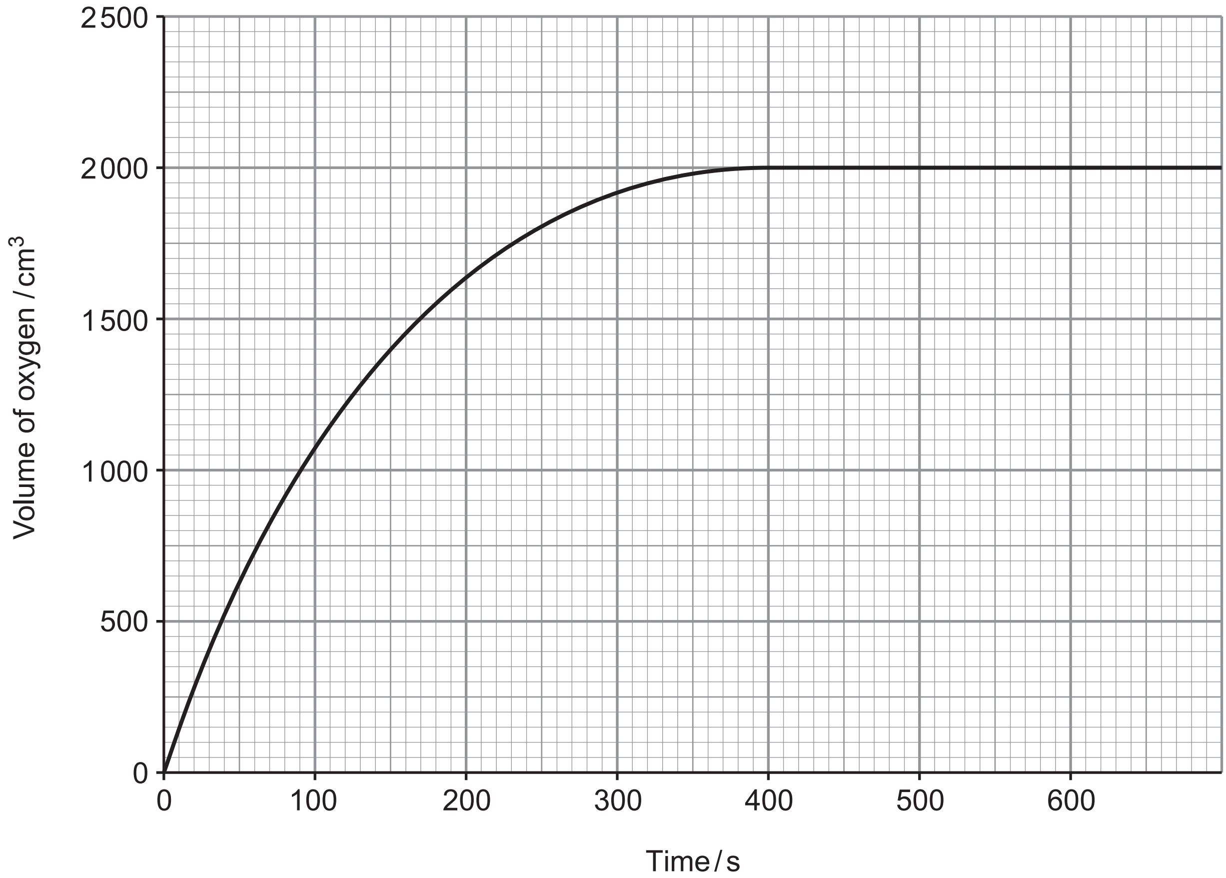
9. A student was investigating the rate of reaction for the catalysed decomposition of hydrogen peroxide to form oxygen.

$$2H_2O_2(aq) \longrightarrow O_2(g) + 2H_2O(l)$$

The student decided to follow the reaction by measuring the volume of oxygen produced at fixed time intervals. He added a spatula measure of manganese(IV) oxide to 100 cm^3 of aqueous hydrogen peroxide solution and started the timer.

(a) Draw a labelled diagram of the apparatus he could use to carry out the experiment and measure the volume of oxygen produced. [2]

(b) The student plotted his results to produce a graph as shown.



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(i) Use the graph to calculate the initial rate of the reaction. Show clearly, on your graph, how you obtained your answer. [2]

Initial rate = ... $cm^3\ s^{-1}$

(ii) Use the graph to calculate the rate of the reaction at 200 seconds. Show clearly, on your graph, how you obtained your answer. [1]

Rate = ... $cm^3\ s^{-1}$

(iii) Use collision theory to explain why the rates you have calculated in parts (i) and (ii) are different. [2]

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(iv) Use the graph to calculate the concentration, in mol dm^{-3}, of the hydrogen peroxide solution used by the student. Give your answer correct to **three** significant figures. [3]

Assume that the reaction was carried out at 298 K and 1 atm pressure.

Concentration = ... mol dm^{-3}

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(c) A second student said that she had seen cobalt(II) chloride speed up the rate of production of oxygen very significantly in a different reaction.

Describe how the students could determine if cobalt(II) chloride is a better catalyst than manganese(IV) oxide in the decomposition of hydrogen peroxide. [3]

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Examiner only

9. *(a)* (i) The average bond enthalpy of a C—C bond is quoted as 348 kJ mol^{-1}.

Explain what is meant by *bond enthalpy*. [2]

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(ii) Ethyne, C_2H_2, contains a C≡C. It reacts with hydrogen in a similar way to ethene.

$$H{-}C{\equiv}C{-}H \quad + \quad 2H_2 \longrightarrow C_2H_6$$

Some average bond enthalpies are given in the table.

Bond	Average bond enthalpy / kJ mol^{-1}
C≡C	839
C—C	348
C—H	413
H—H	436

Use the data to calculate the enthalpy change, Δ*H*, for the reaction of ethyne and hydrogen. [3]

Δ*H* = ... kJ mol^{-1}

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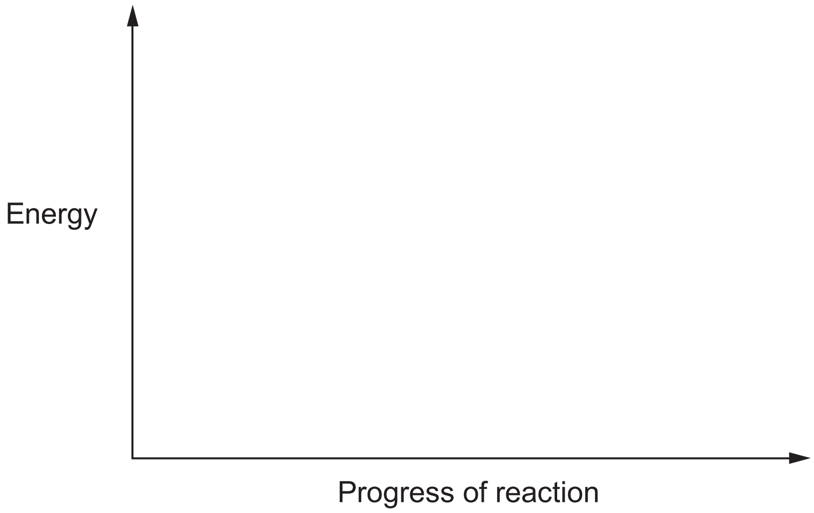
(iii) Use your answer to part (ii) to sketch an energy profile diagram for this reaction on the axes below. Label ΔH and the activation energy, E_a, on your diagram. [2]



(b) Enthalpy changes of reaction are often found indirectly. The enthalpy change for the reaction of ethyne with hydrogen, as shown in part *(a)*, can be determined by using enthalpy changes of combustion.

The table gives some enthalpy changes of combustion, $\Delta_c H^\theta$.

Substance	Enthalpy change of combustion, $\Delta_c H^\theta$ / kJ mol^{-1}
hydrogen, H_2	–286
ethyne, C_2H_2	–1300
ethane, C_2H_6	–1600

Use these enthalpy changes to calculate the enthalpy change, ΔH, for the reaction of ethyne and hydrogen. [3]

$$C_2H_2 \quad + \quad 2H_2 \longrightarrow C_2H_6$$

ΔH = ... kJ mol^{-1}

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(c) The theoretical values that you have calculated in parts *(a)*(ii) and *(b)* are both the enthalpy change for the reaction between ethyne and hydrogen.

Suggest a reason why these values are not the same. [1]

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(d) Suggest the type of reaction that occurs between ethyne and hydrogen. [1]

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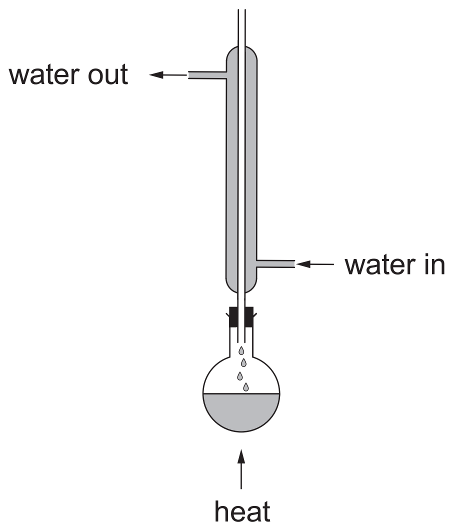
9. *(a)* Alcohols react with carboxylic acids to form esters. To prepare a pure sample of an ester a condenser can be used in the first stage.



(i) Name the method being used with the condenser positioned in this way. Explain why it is necessary to use the condenser. [2]

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(ii) Name the method used to separate the product from the liquid mixture. State which property of an ester allows this method of separation to be used. [2]

Method ....

Property ....

(iii) Name the catalyst most commonly used in the preparation of an ester. [1]

....

(iv) Write the equation for the reaction between methanoic acid and butan-2-ol. Clearly show the structure of the ester formed. [2]

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(b) A compound is known to be one of butan-2-ol, $CH_3CH_2CH(OH)CH_3$, 2-methylpropanoic acid, $CH_3CH(CH_3)COOH$, and 3-hydroxybutanoic acid, $CH_3CH(OH)CH_2COOH$.

Choose **two** chemical tests that will enable you to determine which one it is.

Complete the table below.

Give the reagent(s) for each test and the observation expected for a positive result.

For each test put a **tick (✓)** in the box to show the compound that gives a positive result and a **cross (×)** for those that do not. [5]

Reagent(s)	Observation expected for positive result	butan-2-ol	2-methylpropanoic acid	3-hydroxybutanoic acid

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Examiner only

9. Chloe was investigating the effect of using catalysts on the rate of reaction.

She added 50 cm^3 of 0.1 mol dm^{-3} iron(III) nitrate solution to 50 cm^3 of 0.2 mol dm^{-3} sodium thiosulfate solution. The reaction forms a deep violet iron(III) complex which is unstable and is gradually reduced to form a light green iron(II) complex.

Chloe monitored the rate of reaction by measuring the absorption of light at a wavelength of 500 nm every 10 seconds for two minutes using a data logger.

(a) The violet complex appears black at the beginning of the reaction. State the name of the technique used to monitor the rate of reaction by measuring the absorption of light. [1]

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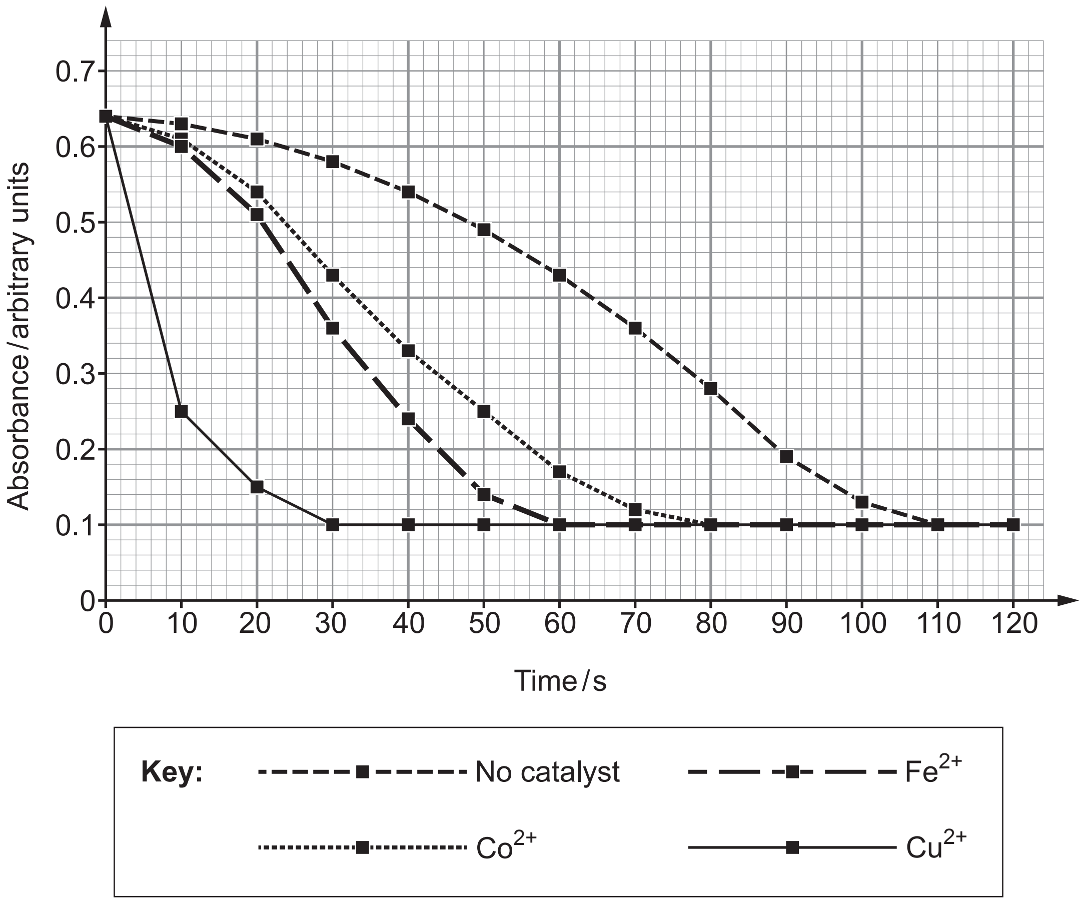
(b) Chloe repeated the experiment three times adding 1 cm^3 of a different catalyst each time at a concentration 0.10 mol dm^{-3}. Below is a graph showing her results:



(i) State which catalyst is the most effective. Explain your answer. [2]

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(ii) Calculate the initial rate of reaction for the reaction catalysed by the copper(II) ions. [2]

rate = ... s^{-1}

(iii) Each catalysed reaction contained the same number of moles of catalyst at the beginning of the reaction. Calculate the moles of catalyst left at the end of the reaction. [1]

moles = ... mol

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(c) Increasing temperature and the addition of a catalyst are two ways of increasing the rate of a reaction.

Using your knowledge of the Boltzmann distribution and particle theory, explain how the rate of reaction is increased using these two different methods.
You may use a diagram(s) to support your answer. [6 QER]

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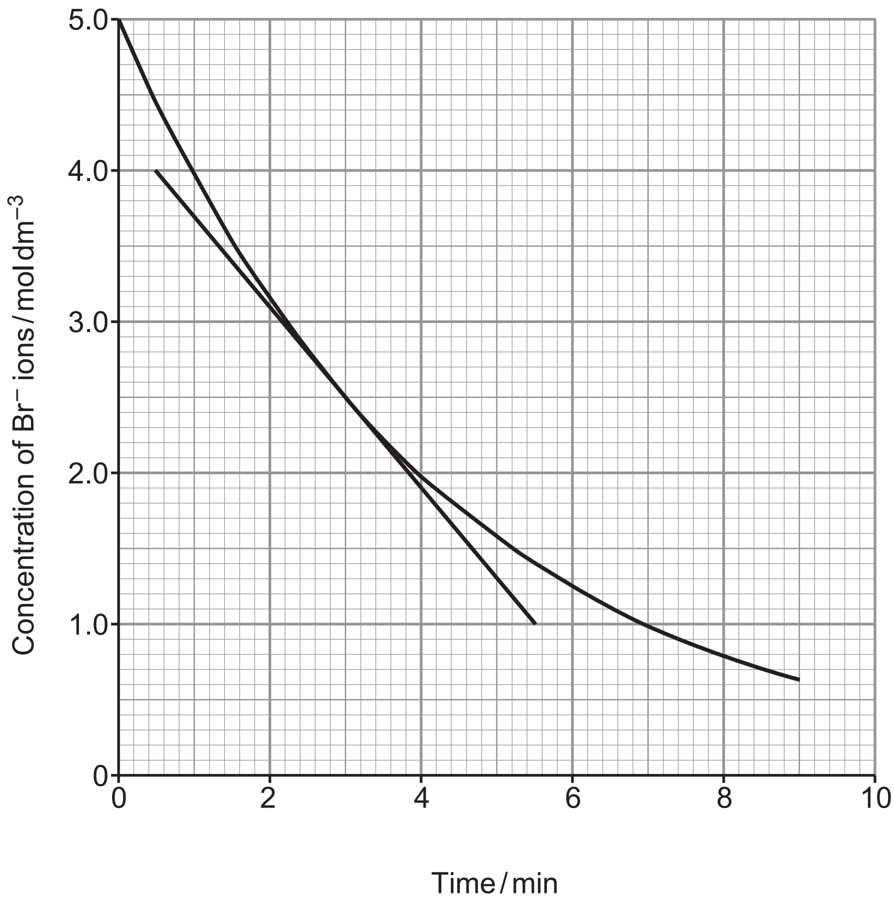
9. Bromide ions react with bromate(V) ions, BrO_3^-, as shown in the equation.

$$5Br^-(aq) \; + \; BrO_3^-(aq) \; + \; 6H^+(aq) \longrightarrow 3Br_2(aq) \; + \; 3H_2O(l)$$

Some students investigated the rate of this reaction by following the concentration of bromide ions over time. They plotted their results as shown.



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(a) (i) The students calculated that the initial rate of the reaction was $1.20\,mol\,dm^{-3}\,min^{-1}$.

Use the tangent to the curve to calculate the rate of the reaction when the concentration of bromide ions had fallen to $2.50\,mol\,dm^{-3}$.

You **must** show your working. [2]

Rate = ... $mol\,dm^{-3}\,min^{-1}$

(ii) Use your answer to part (i) to suggest the relationship between the rate of this reaction and the concentration of bromide ions. [1]

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(b) Suggest how the students could follow the rate of this reaction. Explain your answer. [2]

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(c) In the experiment, the concentration of bromide ions fell from $5.0\,mol\,dm^{-3}$ to $2.0\,mol\,dm^{-3}$ over the first 4 minutes. The initial concentration of bromate(V) ions was $1.0\,mol\,dm^{-3}$.

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3Br_2(aq) + 3H_2O(l)$$

Using the equation, deduce the concentration of bromate(V) ions after 4 minutes. [1]

Concentration of bromate(V) ions = ... $mol\,dm^{-3}$

6